What Mathematical Work Do AI Tutors Leave for Learners? A Qualitative Study and a Test of Decision-Guided Tutoring

Jingying He, Jiawei Ni, Peilin Ying, Duc Le, and Shayan Ahmad
University of Pennsylvania

Abstract: AI tutors must help students without completing the mathematical reasoning they need to practice. Using TutorMoments-Preview, a dataset of real tutoring conversations, we examine this problem in two ways. First, we code the work left for learners in 38 moments. Explicit instructions about balancing support and challenge produced more replies inviting connections, explanations, or strategies than general tutoring guidance. Second, we test whether a separate model can improve a tutor by choosing scaffolding or rigor before the tutor replies. We compare Jev and OpenAI Decisions with a tutor choosing its own approach and with fixed instructions across 520 moments. Automated first-reply pass rates were 68.7% for self-choice, 72.9% with Jev, 73.5% with Decisions, and 74.0% with rigor instructions at every moment. The results show modest gains over self-choice but no advantage over the strongest simple control, highlighting the distinction between selecting a teaching direction and delivering appropriate support.

Introduction

A mathematics tutor must decide when to help and when to leave the next step to the learner. *Scaffolding* makes a task accessible, for example through a hint or representation. *Pushing for rigor* asks the learner to explain a relationship, justify a step, or choose a strategy. Too much help removes reasoning the student needs to practice; too little can leave the student stuck. This assistance dilemma is central to tutoring-system design (Koedinger & Aleven, 2007). Research on teacher questioning likewise emphasizes the mathematical thinking elicited from students (Franke et al., 2009).

Using TutorMoments-Preview, a public collection of tutoring conversations and AI continuations (Zhang et al., 2026), we examine this problem in two ways. First, we code what mathematical work remains for the learner after a tutor responds. Second, we test whether a separate model can choose the teaching approach before another model writes the reply. The motivation is to give a less expensive tutor a clear instructional direction, while checking whether that extra decision actually improves its response.

Separating instructional planning from language generation has precedents. StratL guides tutors through teaching intents (Puech et al., 2025); DICE distinguishes whether to intervene from the ability to answer (Pal et al., 2026). Eduardo instead trains smaller tutors using feedback on tutoring exchanges (Macina et al., 2026). We test an additional decision step without retraining. Our questions are: **(1)** How do tutoring instructions change the mathematical work left for learners? **(2)** Does a separate scaffold-or-rigor decision improve tutor replies beyond detailed instructions alone, and what does it add in cost and delay?

What TutorMoments contains and how we use it

TutorMoments-Preview contains 462 de-identified transcripts of remote tutoring with students in grades 2–7. Transcripts include dialogue and descriptions of shared-screen activity. A *moment* is a stopping point where the tutor must decide what to do next. Experienced teachers described important situations, the human tutor's action, and what followed. The dataset authors used language models to map these descriptions to categories and combined annotations by majority vote. Their evaluation set contains 520 moments, balanced between 260 scaffolding and 260 rigor opportunities (Zhang et al., 2026).

The original benchmark asks an AI tutor to continue each conversation with a simulated learner. Its *plain prompt* gives general tutoring guidance. Its *evaluation-aware prompt* explains scaffolding, rigor, and over-scaffolding, and asks the tutor to respond appropriately without revealing the assigned category. Explicit guidance improved performance, but models still tended to give too much help and miss opportunities for challenge. The scoring can credit responses combining scaffolding and rigor. The labels therefore describe instructional opportunities, not mutually exclusive forms of good teaching.

Our qualitative study uses released human and AI continuations. Our technical study generates new tutor replies from the same histories. We examine the first response, not the original multi-turn exchange. Our scores consequently measure a different outcome from the published leaderboard.

Study 1: What work does the tutor leave for the learner?

We held the AI model constant, comparing GPT-5.5 under the plain and evaluation-aware prompts, alongside the observed human continuation. We randomly selected 40 mathematics moments from 226 eligible situations after excluding students used in earlier pilots: one moment per student and 20 per category. Two were excluded because additional learner activity occurred before the human reply, making it unsuitable for comparison with the AI reply at the selected stopping point. The final sample contained 38 moments, 19 per category, and 114 responses.

The unit was the first *tutor response block*: consecutive tutor utterances ending when the learner next spoke or acted. Following a theory-informed content-analysis approach (Hsieh & Shannon, 2005), we developed two codes. *Work Left* records the most demanding mathematical contribution requested but not already supplied: no new work or repetition (W0), calculation or heavily cued recognition (W1), connection or explanation within a narrowed path (W2), or strategy, argument, or diagnosis with meaningful discretion (W3). *Information Supplied* records how much of the current answer the tutor gives: none (R0), general guidance (R1), a specific step or relationship (R2), or the complete answer (R3). NA means the demand or answer information was unclear. These codes describe the response, not whether its demand suits the learner.

Two separate AI coding sessions applied the same codebook to responses with their human/AI identities hidden. Each had preceding dialogue and access to the full history, and recorded a code, supporting quotation, and rationale. We report the first pass and retain the second to examine interpretive differences. All 114 texts and evidence excerpts were checked against the source records. The main comparison is between the 38 pairs of AI replies. Human continuations provide a naturalistic reference rather than a controlled prompt comparison.

Qualitative findings

Evaluation-aware replies more often left a connection, explanation, or strategy for the learner: 16/38 received W2 or W3, compared with 2/38 plain replies (Table 1). Plain replies supplied the complete active answer in 21/38 cases, versus 5/38 evaluation-aware replies; one plain reply was unclear on this measure.

Table 1: Mathematical work left in the first coding pass ($n = 38$ per column).

Work left	Plain	Evaluation-aware	Human reference
W0: No new work or repetition	21	2	20
W1: Calculation or cued recognition	15	20	12
W2: Connection or explanation	2	14	1
W3: Strategy, argument, or diagnosis	0	2	0
NA: Unclear demand	0	0	5

Three cases from the accompanying casebook show what these codes capture. In E01, the task was $25 \times 14 = 25 \times (2 \times \star)$. The plain reply supplied 7 and verified the equality (W0, R3). The evaluation-aware reply asked, "What number times 2 makes 14?" It supplied the relationship but left the calculation (W1, R2). A question alone therefore did not indicate substantial reasoning.

In E06, both replies rewrote $5/7 - 3/7$ as $(5 - 3)/7$. The plain reply asked for $5 - 3$ and the final answer (W1, R2). The evaluation-aware reply additionally asked, "And why do we keep the denominator as 7?" The subtraction rule was supplied, but its mathematical explanation remained for the learner (W2, R2).

In E11, both replies gave 17% as the conversion of 0.17 (R3). The plain reply only asked the learner to select that answer (W0). The evaluation-aware reply also asked why it was not 170% (W2). Giving the numerical answer can therefore coexist with leaving a new explanation to the learner. This is why we coded supplied information and remaining work separately.

The second pass also favored evaluation-aware replies, assigning W2/W3 to 23 rather than five plain replies. Different counts show that distinguishing calculation from explanation requires judgment; the examples illustrate coding decisions, not measured student learning.

Study 2: Can a separate decision improve the next tutor reply?

What a decision model does

Here, a *decision model* reads the conversation and selects one of two predefined actions, scaffolding or rigor. It does not write the message to the student. A separate language model then writes that message using the chosen teaching instructions. An ordinary language model can also make this choice; our question is whether assigning it to a separate system improves the reply.

TypeSafe introduced Jev as a "System One" model designed to return fast, structured choices and probabilities rather than conversational text (Almeida, 2026). We used Jev 1.14.0 and OpenAI's Decisions API, a software interface for requesting choices from GPT-6 Luna (OpenAI, n.d.). Qwen3-32B wrote the tutor replies (Qwen Team, 2025). These are distinct roles: Jev or Decisions selects an approach; Qwen turns it into language.

Figure 1 shows the proposed separation. We wrote two instruction sets. Scaffolding asks for the smallest useful hint while leaving the key calculation to the student. Rigor asks for a justification, connection, or independent step without first giving the reasoning. Both require a response appropriate to the current task. The experiment tests whether choosing between them adds value beyond good instructions alone.

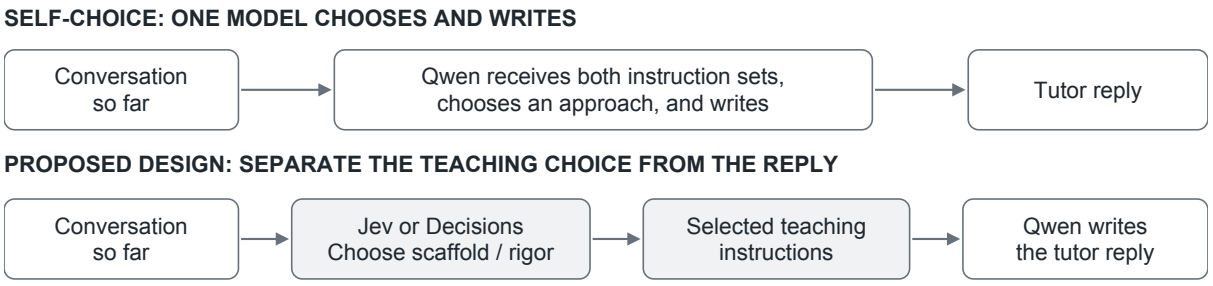


Figure 1: Proposed decision-guided tutoring and our controlled test. The decision step chooses instructions; Qwen writes the reply using the full conversation.

How we tested the design

First, we tested category selection across all 520 moments. The selected prompts matched the assigned label in 376 cases for Jev (72.3%) and 386 for Decisions (74.2%). These saved predictions were used in the response experiment. Prompt development had already used this dataset, so the results concern previously inspected cases. Later prompt revisions reduced agreement for both systems; all versions are retained.

Second, Qwen generated three replies per moment: **self-choice**, with both detailed instruction sets and a request to choose; **scaffold-guided**, with scaffolding instructions; and **rigor-guided**, with rigor instructions. All retained the publisher's evaluation-aware prompt. This produced 1,560 replies. A Jev or Decisions prediction selected the matching guided reply. Reusing the same candidates isolates the effect of selecting an instruction from differences between repeated generations.

Two controls always selected the scaffold-guided reply or always selected the rigor-guided reply, regardless of the assigned category. They test whether the extra decision step beats a consistent instruction. The instruction name does not guarantee the behavior produced: rigor guidance still allowed context-sensitive support. A diagnostic selected using the dataset label, which neither decision system received.

All models received the complete preceding conversation, excluding future turns and teacher annotations. Qwen also retained the publisher's tutoring-platform identifier, the same value in all 520 cases. All conditions used the same generation settings and response-length limit; no reply was cut off. Exact prompts and settings are in the supporting files.

We also rechecked results for 346 moments retained during earlier data preparation for the coding study. It retained mathematics tasks, removed repeated stopping points, and required usable, aligned responses from the human tutor and all 14 original AI conditions. This additional check applies more restrictions than subject matter alone; it is separate from the 38 moments we coded.

How replies were evaluated

Gemini 2.5 Flash, a separate AI evaluator, rated the three anonymous replies in randomized order, without condition names or dataset labels. It assessed scaffolding, rigor, over-helping, mathematical errors, and fit to the conversation. We then compared those judgments with the assigned category. Passing required the relevant behavior, no over-helping or mathematical error, and full context fit. A reply containing both scaffolding and rigor

could satisfy either label.

Claude Sonnet 4.6 repeated the ratings for 32 moments, one per student and balanced by category. Comparisons paired replies from the same moment. We estimated uncertainty by repeatedly resampling students, keeping each student's moments together. This avoids treating several moments from one student as independent observations.

Technical findings

Jev-guided replies passed in 379/520 cases and Decisions-guided replies in 382, compared with 357 for self-choice (Table 2). These were gains of 4.2 percentage points for Jev (95% confidence interval: -1.0 to 9.2) and 4.8 points for Decisions (0.2 to 9.5). However, always using rigor guidance passed 385 cases, slightly more than either decision system. Jev improved performance on rigor opportunities from 149 to 180 passes, while reducing scaffolding passes from 208 to 199.

Table 2: Passing first replies. Each row uses the same 520 conversation stopping points.

How instructions are selected	All 520	Scaffold 260	Rigor 260	Math set 346
Qwen chooses its approach	68.7%	80.0%	57.3%	72.0%
Jev chooses	72.9%	76.5%	69.2%	72.8%
OpenAI Decisions chooses	73.5%	80.8%	66.2%	75.1%
Always use scaffold instructions	56.5%	86.2%	26.9%	59.8%
Always use rigor instructions	74.0%	65.8%	82.3%	76.0%
Use the dataset label (diagnostic)	84.2%	86.2%	82.3%	86.4%

The 346-case set applies the additional selection checks described in the methods. The diagnostic uses the assigned label, which an operating tutor would not have. Scores concern our first-reply evaluation.

In the 346-moment comparison set, Jev's advantage over self-choice was only 3 cases, or 0.9 points (interval -5.3 to 6.5). Evaluation was also sensitive to the judge. On the same 32 cases, Gemini passed 20 self-choice and 19 Jev replies; Claude passed 16 and 22, respectively. The two evaluators agreed on 75 of the 96 individual reply judgments (78.1%).

Correct category selection did not guarantee a good response. Of Jev's 376 category matches, 320 selected replies passed; 59 of its 144 mismatches also passed. In one case, both decision systems correctly selected scaffolding for a learner working on $1{,}000 - 10$, but Qwen's reply returned to an earlier $338 + 586$ problem. Even selecting by the dataset label passed only 438/520 cases. The remaining failures concern how the tutor carries out its instruction, not just which instruction is chosen.

We found no speed advantage. Qwen's median time to begin answering was 1.20 seconds for self-choice. Adding the separately measured decision durations gave estimated totals of 1.43 seconds with Jev and 1.60 with Decisions. These are reconstructed estimates, because saved decisions selected pre-generated replies; we did not time a live end-to-end system. Estimated model costs per 1,000 replies were \$0.59, \$0.90, and \$1.25, respectively, excluding evaluation.

Discussion and contribution

The qualitative analysis describes what the learner can contribute after a reply. The technical experiment tests whether a separate choice helps produce that reply. Together they distinguish selecting a teaching direction from carrying it out. A model can choose scaffolding correctly yet address the wrong task; a response can also combine support and challenge.

Neither decision system established an advantage over fixed rigor guidance. This comparison shows why an extra decision step should be assessed against equally detailed self-choice and fixed instructions. The fixed-rigor result does not mean every learner needs more challenge: 202 of its 520 replies were judged to contain both scaffolding and rigor. The instruction could encourage independent reasoning while still offering support. A future decision model might choose a concrete next action and information to withhold, rather than only a broad category.

Our contribution is a codebook separating supplied answers from remaining mathematical work, and a controlled test of what a separate decision step adds beyond detailed instructions. These exploratory results use previously inspected cases, AI-assisted coding, and automated ratings. They do not measure student learning: the new AI replies were not delivered to learners. New conversations and independent human evaluation are needed to test classroom benefit. AI tools assisted coding, programming, evaluation, and drafting.

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Supplementary materials

Complete study materials on Google Drive: paper, data, and analysis files, viewable without sign-in.

Qualitative analysis: coding manual, workbook, all 38 cases, both coding passes, and selection records.

Technical analysis: prompts, all 520 decisions and response results, evaluator records, and runnable code.

GitHub repository: maintained implementation and results. The repository is private; the technical Drive folder provides the complete review package without GitHub access.